

Soft Actor-Critic (SAC) vs Linear Programming (LP) for battery arbitrage in a local energy market



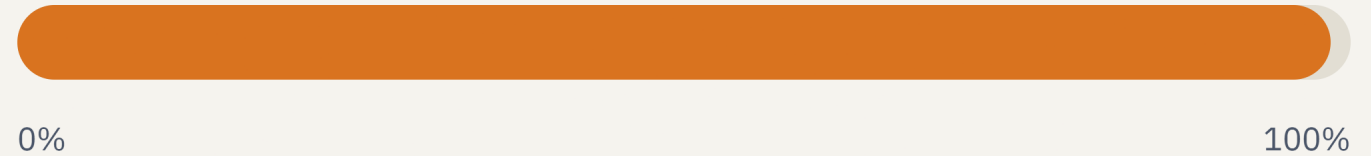
SAC learns

LP calculates

THE SHORT ANSWER

The RL agent earned 98% of the optimiser's profit

98%



€603 vs €612 over 89 days

Two ways to drive the same battery



SAC learns

- Trial and error over 300,000 training steps
- Outputs one number: how hard to charge or discharge



OP calculates

- Solves the best plan for the next 6 hours
- Acts for 15 minutes, then re-plans
- Provably optimal inside its window



Both see the same thing: current price + next 8 h of day-ahead prices

The early "3× win" was unfair

Profit

€634



SAC

€201



Optimiser

Profit per kWh

2.24 ct



SAC

2.22 ct



Optimiser

Steps where it bought

46% vs 7%

The OP was also managing all the other nodes.

22 TO 23 AUGUST

The rematch: a level playing field

✓	Information	Current price + next 6h of day-ahead prices	Same
✓	Goal	Trading cash + value of energy left at the end	Same
✓	Battery physics	Shared code	Same
✓	Order placement	Same bid and ask function and safety rules	Same
⚡	How it decides	Learned network vs exact optimisation	Different

SAC vs OP · 05 · Both ran all 89 days on the same code version

HOW THEY TRADE

Two trading styles

SAC

Many medium orders. Holds charge for bigger spreads.



40% left at the end

8 kWh

per order

80

fills a day

2.2

cycles a day

OP

Full power or nothing. Runs its inventory down.



7% left at the end

18 kWh

per order (max)

47

fills a day

2.8

cycles a day

FINAL RESULT, Q1 2023

Same profit, different route

■ SAC ■ Arbitrage LP

Profit (incl. leftover charge)

SAC = 98.5% of LP



Energy sold

LP moved 29% more



Profit per kWh sold

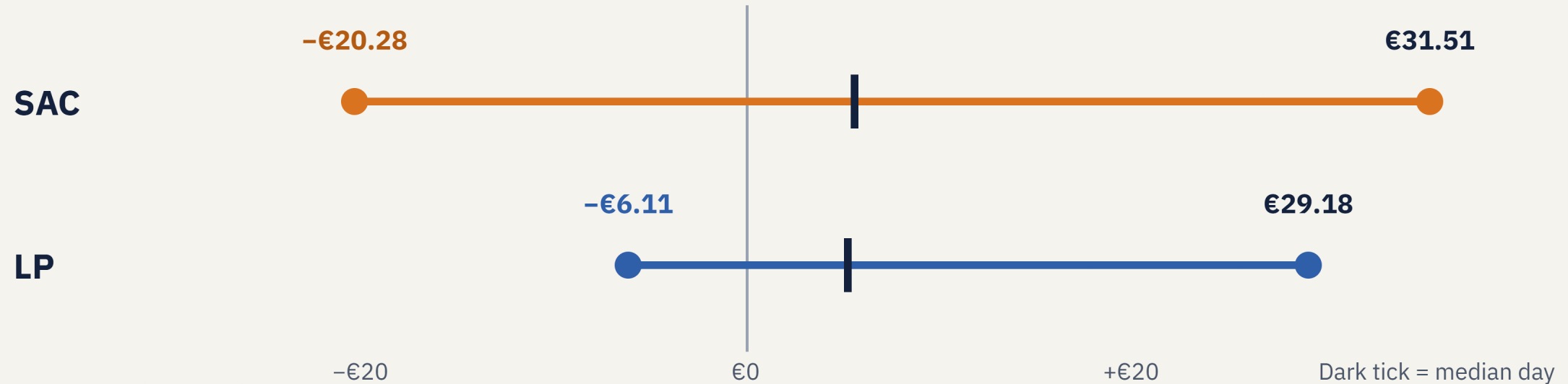
SAC earns 26% more



SAC vs OP · 07 · The 1.5% profit gap is inside the ±1.5% measurement noise

DAY BY DAY

The OP has much better bad days



Losing days **17 vs 16**

Median day **€5.63 vs €5.31**

SAC won **41 of 89 days**

SAC vs OP · 08 · Range of daily profit, worst to best day

BATTERY WEAR

Price battery wear, and SAC pulls ahead



AT 1 CT PER KWH

€315 vs **€246**

SAC earns 28% more because it cycles the battery less.

SAC vs OP · 09 · Wear subtracted after the fact.

SCORECARD

Who won? It depends what you count

Total profit, wear ignored	LP +1.5%, inside the noise	Tie
Profit per kWh cycled	+26%	SAC
Profit with any wear cost	+28% at 1 ct per kWh	SAC
Median day	€5.63 vs €5.31, slightly	SAC
Worst day and risk	-€6 vs -€20, clearly	LP
Guarantees and explainability	Can prove its plan is optimal	LP

LIMITS

What no fix can remove



±1.5% noise

Each controller needs its own market run, and the battery moves local prices.



Perfect forecasts

Both see the real prices ahead.



No guarantee for SAC

Its performance is measured, never proven.
Unseen markets can break it.



Thin margins

Only about 2 ct per kWh is on the table. No controller can create spread.

VERDICT

A learned policy matched an exact optimiser, and got more out of every kWh.



Pick the LP when worst case and guarantees matter



Pick SAC when battery wear matters